Blueprint for PINN-Based Nonlinear Meta-Impactor Dispersion Prediction

R. Zhang V. Dertimanis M. Haywood–Alexander E. Chatzi

March 20, 2026

1 Introduction

This blueprint outlines a two-stage framework for predicting the dispersion characteristics of nonlinear meta-impactor lattices using Physics-Informed Neural Networks (PINNs). Unlike classical Bloch-based analyses, which rely on harmonic assumptions and predefined impact cycles, the proposed method learns the nonlinear operator directly from time-domain data, enabling predictions for arbitrary lattice configurations without additional simulations.

2 Stage 1: Training the Nonlinear Operator

2.1 Objective

The first stage aims to train a PINN to learn the operator that maps the lattice parameters, excitation characteristics, and number of unit cells to the system response. The network captures the nonsmooth dynamics due to repeated impacts in each unit cell.

2.2 Inputs and Outputs

- Inputs: Unit cell parameters (m, μ, k, c, D, r), number of unit cells N, and excitation signal.
- Outputs: Time-resolved system response of all masses, including both primary and internal masses.

2.3 Training Strategy

1. Generate reference data for a set of representative parameters and excitations using the sequential PINN framework.
2. Optimize the PINN parameters to minimize the physics-informed loss, which includes the governing equations, initial conditions, and impact constraints.

3. Validate the trained operator by comparing predicted responses to event-based simulations.

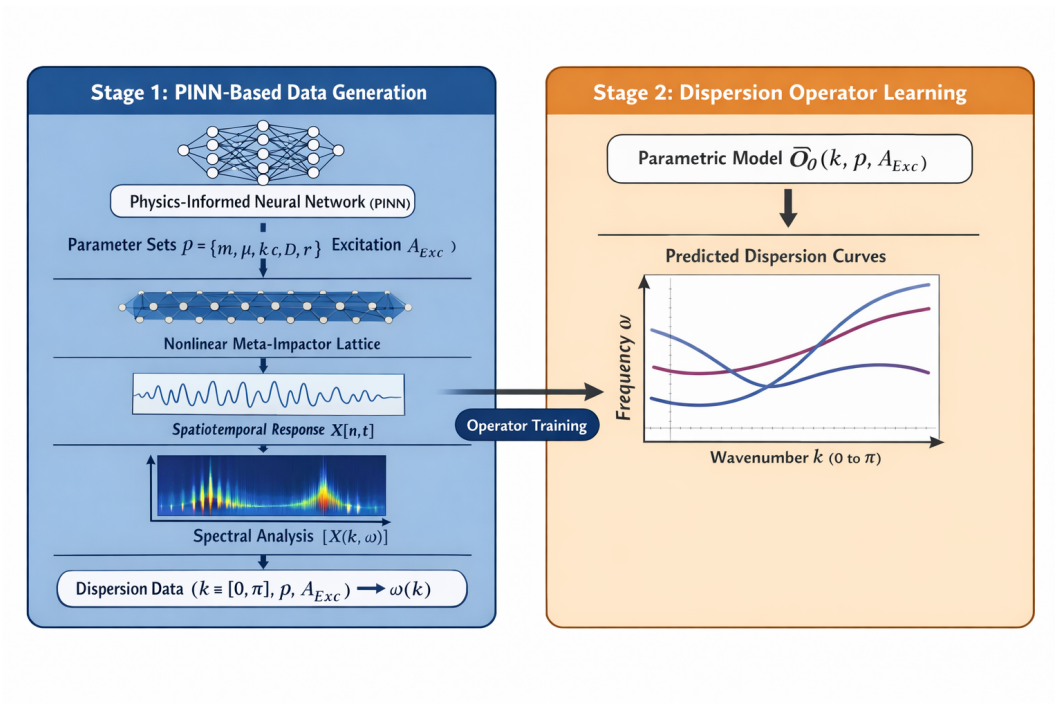


Figure 1: Stage 1: Training the nonlinear operator from representative lattices. Stage 2: Using the trained operator to predict dispersion for arbitrary lattices without simulation.

3 Stage 2: Dispersion Prediction for Arbitrary Lattices

3.1 Objective

Once the operator is trained, Stage 2 enables predicting the dispersion relation for any lattice configuration, without performing new time-domain simulations. This includes lattices with arbitrary numbers of unit cells and parameter sets.

3.2 Inputs and Outputs

- Inputs: Trained operator, number of unit cells N, lattice parameters, and a given excitation.
- Outputs: Schematic or quantitative frequency–wavenumber maps (dispersion curves), including multi-band effects for finite lattices.

3.3 Periodic vs. Finite Lattice

- **Periodic (Bloch) Lattices:** Classical Bloch analysis assumes infinite repetition and harmonic waves. Frequency bands are obtained analytically but rely on assumptions about the number of impacts per cycle, which may be invalid for strongly nonlinear lattices.
- **Finite Lattices (PINN-based):** The PINN-trained operator allows the computation of dispersion relations for lattices of arbitrary length, including the effects of finite size, boundary conditions, and nonsmooth dynamics.

3.4 Bloch vs. PINNs

- **Bloch Analysis:** Assumes harmonic solutions and predefined impact sequences. Valid for linear or weakly nonlinear lattices but cannot capture arbitrary multi-impact responses.
- **PINN Approach:** Does not require a harmonic ansatz. Learns the underlying nonlinear operator from representative time-domain responses. Can generalize to new lattices and excitations, predicting the frequency-wavenumber relation directly.

4 Schematic Predicted Dispersion Curves for Different Lattices

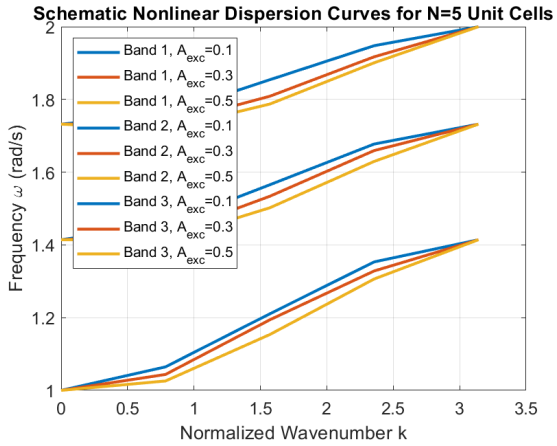
Figure 2 illustrates the predicted nonlinear dispersion curves for finite meta-impactor lattices with varying numbers of unit cells and excitation amplitudes, using the trained PINN operator.

Notes:

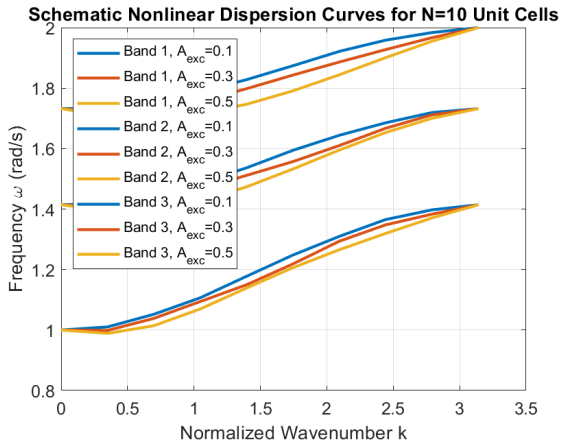
- The number of unit cells N affects the discretization of the wavenumber axis and introduces finite-lattice effects in the dispersion curves.
- Multi-band structures appear due to the internal masses in the unit cells (acoustic and optical branches).
- Small bumpy features along the curves represent vibro-impact nonlinearities.
- Different excitation amplitudes (colors) indicate the amplitude-dependent frequency shifts and nonlinearity intensity.

5 Conclusion

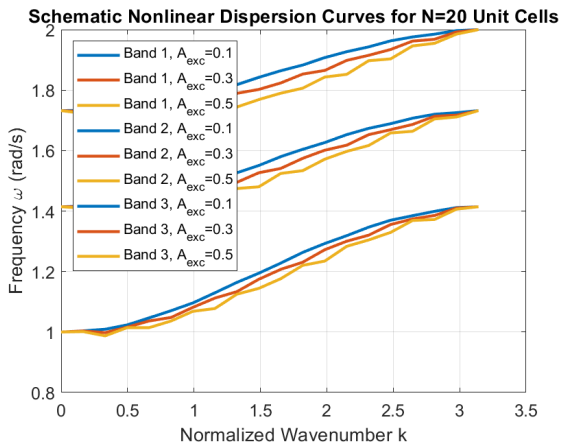
The two-stage framework enables the extraction of nonlinear dispersion characteristics for meta-impactor lattices using PINNs. Stage 1 trains the operator from representative lattices, while Stage 2 leverages the trained operator to predict dispersion curves for arbitrary lattice



(a) Small lattice, $N = 5$ unit cells.



(b) Medium lattice, $N = 10$ unit cells.



(c) Large lattice, $N = 20$ unit cells.

Figure 2: PINN-predicted nonlinear dispersion curves for meta-impactor lattices with varying numbers of unit cells and excitation amplitudes. Multi-band effects and impact-induced nonlinear fluctuations are visible.

configurations without performing new simulations. The approach accounts for both finite-lattice effects and impact-induced nonlinearities, overcoming the limitations of classical Bloch analysis.